

Joanna Luc

PERSONAL DETAILS

Affiliation Institute of Philosophy, Jagiellonian University, Kraków, Poland (position: adiunkt)
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RESEARCH INTERESTS

Philosophy of physics, general philosophy of science, metaphysics, epistemology.

EDUCATION – DEGREES

2020 **PhD in Philosophy**
Jagiellonian University, Kraków, Poland
Dissertation title: “Modalities in the Physical World”
Supervisor: Prof. Tomasz Placek
Reviewers: Prof. Tomasz Bigaj, Prof. Barbara Vetter

2019 **MPhil in History and Philosophy of Science and Medicine**
Cambridge University, Cambridge, United Kingdom

2018 **MSc in Physics**
Jagiellonian University, Kraków, Poland

2016 **MA in Philosophy**
Jagiellonian University, Kraków, Poland

2016 **BSc in Physics**
Jagiellonian University, Kraków, Poland

2013 **BA in Philosophy**
Jagiellonian University, Kraków, Poland

SELECTED SCHOLARSHIPS AND AWARDS

2021 Polish Prime Minister Prize for PhD Dissertation

2021 START scholarship by Foundation for Polish Science

2018-2019 AHRC scholarship and Santander scholarship for MPhil studies at Cambridge University

2015–2016 Polish Minister of Science and Higher Education Scholarship for Academic Achievements

PUBLICATIONS IN ENGLISH

- 2023 “Can Dispositions Replace Laws in the Description of the Physical World?”, *Journal for General Philosophy of Science*, DOI: 10.1007/s10838-023-09657-2.
- 2023 “Motivationalism vs. interpretationalism about symmetries: some options overlooked in the debate about the relationship between symmetries and physical equivalence”, *European Journal for Philosophy of Science*, 13:40, DOI: 10.1007/s13194-023-00539-4.
- 2023 “The unmeasurability of absolute velocities from the point of view of epistemological internalism”, *Erkenntnis*, DOI: 10.1007/s10670-023-00679-2.
- 2023 “States vs. Changes of States: A Reformulation of the Ontic vs. Epistemic Distinction in Quantum Mechanics”, *Foundations of Physics* 53(22), pp. 1-35, DOI: 10.1007/s10701-022-00662-x.
- 2022 “Arguments from scientific practice in the debate about the physical equivalence of symmetry-related models”, *Synthese* 200:72, DOI: 10.1007/s11229-022-03618-w.
- 2020 “Quantumness of States and Unitary Operations”, *Foundations of Physics* 50(11), pp. 1645-1685, DOI: 10.1007/s10701-020-00391-z.
- 2020 “Modality in Physics”, *Foundations of Physics* 50(6), pp. 515-521, DOI: 10.1007/s10701-020-00344-6, an editorial to Special Issue of *Foundations of Physics* (with Gábor Hofer-Szabó and Tomasz Placek).
- 2020 “Interpreting non-Hausdorff (Generalized) Manifolds in General Relativity”, *Philosophy of Science* 87(1), pp. 21-42. DOI: 10.1086/706116 (with Tomasz Placek).
- 2019 “Generalised manifolds as basic objects of General Relativity”, *Foundations of Physics* 50(6), pp. 621-643. DOI: 10.1007/s10701-019-00292-w.
- 2017 “Slow-roll approximation in loop quantum cosmology”, *Journal of Cosmology and Astroparticle Physics* 45(1). DOI: 10.1088/1475-7516/2017/01/045 (with Jakub Mielczarek).

PUBLICATIONS IN POLISH

- 2019 “Co mówi o prawach przyrody geneza tego pojęcia? Polemika z Ronaldem N. Gierem” [What does the genesis of the notion of laws of nature tell about it? A polemic with R. N. Giere], *Filozofia Nauki* 27(3), pp. 103-116. DOI: 10.14394/filnau.2019.0020
- 2019 “Prawo czy forma? Francisca Bacona koncepcja prawidłowości w przyrodzie” [Law or form? Francis Bacon’s conceptions of regularities in nature], *Filozofia Nauki* 27(1), pp. 95-120. DOI: 10.14394/filnau.2019.0005
- 2018 “Teoria praw przyrody Armstronga wobec problemów identyfikacji i inferencji” [Armstrong’s Theory of the Laws of Nature Confronted with the Problems of Identification and Inference], *Diametros* 55, pp. 132-157, DOI: 10.13153/diam.1180.
- 2017 “Realność problemu zmiany. Polemika z Adamem Andrzejewskim” [Is the problem of change a genuine problem? A polemic with Adam Andrzejewski], *Filozofia Nauki*

25(3), pp. 137-144.

- 2017 “Prawa przyrody jako przedmioty teoretyczne? Krytyczna analiza pomysłu Michaela Tooleya” [Laws of nature as theoretical entities? A critical analysis of Michael Tooley’s ideas], *Filozofia Nauki* 25(1), pp. 49-64.
- 2016 “Cztery współczesne koncepcje kategorii ontologicznych” [Four contemporary theories of ontological categories], *Filozofia Nauki* 24(4), pp. 95-128.
- 2016 “Odpowiedź Lowe’a na argument Ramsey’a przeciwko rozróżnieniu uniwersalia-indywidua” [Lowe’s response to Ramsey’s argument against the distinction between universals-individuals], *Argument: Biannual Philosophical Journal* 6(1), pp. 223-237.
- 2015 “Poznanie metafizyczne w świetle rozróżnień abstrakcyjne-dekonkretne i ogólne-jednostkowe” [Metaphysical knowledge in the light of distinctions: abstract-concrete, universal-particular], *Idea: studia nad strukturą i rozwojem pojęć filozoficznych XXVII*, pp. 215-223.
- 2013 “Zagadnienie redukcjonizmu w filozofii biologii” [The issue of reductionism in the philosophy of biology], *Zagadnienia filozoficzne w nauce* 52, pp. 179-201.
- 2013 “Samowiedza czy samokreacja? O konsekwencjach niefaktualności pytań o siebie dla problemu samowiedzy” [Self-knowledge or self-creation? On the consequences if non-factuality of questions about the self for a problem of self-knowledge], *Studia z Kognitywistyki i Filozofii Umysłu* 7(1), pp. 73-84.

BOOK REVIEWS

- 2020 “W gąszczu filozoficznych rozróżnień: Pawła Rojka systematyzacja podstawowych zagadnień dotyczących tropów i uniwersaliów” [In the maze of philosophical distinctions: Paweł Rojek’s systematization of the basic issues concerning tropes and universals], *Filozofia Nauki* 28(2), pp. 99-111, DOI: 10.14394/filnau.2020.0011. This is a review of Paweł Rojek’s book *Tropy i uniwersalia: Badania ontologiczne* [Tropes and Universals: Ontological Investigations].

RESEARCH GRANTS

- 2023-2026 **Co-investigator**, “Quantum State: Information, Free Choice, Randomness”, Opus 2022/45/B/HS1/02700 (Principal Investigator: Tomasz Placek)
Narodowe Centrum Nauki (National Science Centre), Poland
- 2018–2021 **Principal Investigator**, “Ontological and methodological status of symmetries in physics”, Preludium 2017/25/N/HS1/00705
Narodowe Centrum Nauki (National Science Centre), Poland
- 2017–2020 **Co-investigator**, “Real possibilities: what they are and how they are represented in physics”, Opus 2016/23/B/HS1/00464 (Principal Investigator: Tomasz Placek)
Narodowe Centrum Nauki (National Science Centre), Poland
- 2017–2018 **Principal Investigator**, “Can generalised probabilities explain quantum effects?”
Funding for Young Researchers and Doctoral Candidates, *Faculty of Philosophy, Jagiellonian University*
- 2015–2016 **Co-investigator**, “Cosmological constraints on quantum gravity”, Sonata DEC-

CONTRIBUTED TALKS AT INTERNATIONAL CONFERENCES

- 2.-4.10.2023 “What price relationism in QM?” (with Tomasz Placek)
Perspectivism and Quantum Mechanics, online
- 21.-25.08.2023 “Laws as the best systematizations of facts vs. laws as the best fact-constraining
"algorithms": on the inequivalence of two conceptions of laws of nature”
ECAP11: The 11th European Congress Of Analytic Philosophy, Vienna, Austria
- 8-10.07.2023 “States vs. changes of states: a reformulation of the ontic vs. epistemic distinction in
quantum mechanics” (poster)
Foundations 2023: The 21st European Conference on Foundations of Physics, Bristol,
United Kingdom
- 19.-20.04.2023 “The Unmeasurability of Absolute Velocities from the Point of View of
Epistemological Internalism”
Cambridge Early Career Workshop in the Philosophy of Measurement, online
- 22.-23.04.2022 “On the logic of (relativistic) coordinate frames” (with Tomasz Placek)
Logic4Peace, online
- 21.03.-
25.03.2022 “Arguments from scientific practice in the debate about the physical equivalence of
symmetry-related models”
*Deutsche Physikalische Gesellschaft Conference, Arbeitsgruppe Philosophie der
Physik*, online
- 24-28.08.2020 “Can dispositions replace laws in the description of the physical world?”
ECAP 10 (10 th European Congress of Analytic Philosophy), online
- 6-7.02.2020 “Symmetry and the Equivalence of Models”
Philosophy of Physics Graduate Conference, Oxford, United Kingdom
- 3-4.09.2018 “Is nonlocality a distinctive feature of quantumness?”
*First Irvine-Munich-PoliMi-Salzburg Conference in Philosophy and Foundations of
Physics (IMPS 2018)*, Salzburg, Austria
- 10-13.07.2018 “On generalised probability space representations of quantum mechanical experiments”
(with Tomasz Placek)
Foundations 2018, Utrecht, The Netherlands
- 4-6.07.2018 “Hausdorffness of general relativistic spacetimes” (with Tomasz Placek)
British Society for the Philosophy of Science Annual Conference, Oxford, United
Kingdom
- 11-13.06.2018 “Generalised manifolds as basic objects of General Relativity”
Modality in Physics, Kraków, Poland
- 1-2.03.2018 “What spacetime does?”

Spacetime Functionalism, Geneva, Switzerland

- 6-9.09.2017 “Problems with the notion of energy in General Relativity and physical meaning of this quantity” (poster)
EPSA17 – The Sixth Biennial Conference of the European Philosophy of Science Association, Exeter, United Kingdom
- 23-27.08.2017 “Are non-Hausdorff space-times physically reasonable?”
3rd Logic, Relativity and Beyond Conference, Budapest, Hungary
- 4-7.07.2017 “Why generalised probability does not explain quantum correlations?” (with Tomasz Placek)
IQSA Quantum Structures Workshop, Nijmegen, The Netherlands
- 3-4.07.2017 “Two divisions of symmetries and some of their consequences”
Direct empirical status and the ontology of symmetries in physics, Louvain-la-Neuve, Belgium
- 9-10.06.2017 “Are different laws of nature necessary in different ways?”
Modally-rich metaphysical landscapes, Kraków, Poland
- 11-18.03.2017 “Various Meanings of Symmetries in Physical Theories”
Rethinking Foundations of Physics, Dorfgastein, Austria
- 8-9.10.2016 “Notes on Ehrling’s Conception of the Universal-Particular Distinction”
4th Annual Philosophers’ Cocoon Philosophy Conference, Tampa, United States (skype session)
- 6-12.02.2016 “The construction of slow-roll expansion in Loop Quantum Cosmology”
Quantum Spacetime ’16, Zakopane, Poland
- 9-11.09.2015 “Necessary and accidental qualities in CIFOL”
Entia et nomina, Kraków, Poland
- 15-25.07.2015 “The construction of slow-roll expansion in Loop Quantum Cosmology”
Cosmology School: Introduction to Cosmology, Kielce, Poland

CONTRIBUTED TALKS AT POLISH CONFERENCES

- 11-16.09.2023 “Symetrie i fizyczna równowaga modeli”
12 Polski Zjazd Filozoficzny, Łódź, Poland
- 22-23.11.2017 “Are different laws of nature necessary in different ways?”
Konferencja Filozofii Nauki i Metod Formalnych w Filozofii [Conference on Philosophy of Science and Formal Methods of Philosophy], Lublin, Poland
- 27-28.10.2016 “Problemy z pojęciem energii w ogólnej teorii względności” [Problems with the notion of energy in General Relativity]
Konferencja Filozofii Nauki i Metod Formalnych w Filozofii [Conference on Philosophy of Science and Formal Methods of Philosophy], Kraków, Poland
- 11.05.2016 “Racjonalność naukowa: od rozumienia do przewidywania?” [Scientific rationality: from understanding to predicting?]
Oblicza Racjonalności [Faces of Rationality], Warszawa, Poland
- 4-5.03.2016 “Geneza pojęcia praw przyrody a zagadnienie realizmu naukowego” [The genesis of the notion of laws of nature and the issue of scientific realism]

XII Ogólnopolska Konferencja Filozofii Fizyki: Prawda a Poznawanie Świata [XII Polish Conference in the Philosophy of Physics: Truth and Knowledge about the World], Poznań, Poland

- 26-27.09.2015 “Odpowiedź Lowe’a na Ramsey’a problem uniwersaliów” [Lowe’s response to Ramsey’s problem about universals]
ROCZNICE: Perspektywy analitycznej: filozofii nauki, filozofii języka i fenomenologii [Anniversaries: Perspectives of analytic philosophy of science, language and of phenomenology], Kraków, Poland
- 2-4.07.2015 “Argument na rzecz platonizmu na gruncie koncepcji kategorii Lowe’a” [The argument for Platonism within Lowe’s conception of categories]
Zlot Filozoficzny [Philosophers’ Rally], Lublin, Poland
- 4-8.05.2015 ‘The classification of properties in CIFOL’
Zastosowania Logiki w Filozofii i Podstawach Matematyki [Application of Logic in Philosophy and Foundations of Mathematics], Szklarska Poręba, Poland

TALKS AT SEMINARS ETC.

- 29.06.2023 “The analysis of dispositions in terms of the fundamental laws of nature and normal conditions”
Research Colloquium, University of Konstanz, Konstanz, Germany
- 10.05.2023 “Symmetries and the physical equivalence of models”
Talk in the Epistemology Unit, Jagiellonian University, Kraków, Poland
- 11.01.2022 “Generalised manifolds as basic objects of general relativity”
Western University Philosophy of Physics Reading Group, online
- 18.05.2022 “Laws as the axioms of the best system vs. laws as the best fact-predicting ‘algorithms’: on the inequivalence of two conceptions of laws of nature”
Talk in the Epistemology Unit, Jagiellonian University, Kraków, Poland
- 28.04.2021 “A diachronic perspective on the ontic vs. epistemic interpretation of quantum states”
Talk in the Epistemology Unit, Jagiellonian University, Kraków, Poland
- 15.01.2020 “Laws of nature vs. dispositions”
Talk in the Epistemology Unit, Jagiellonian University, Kraków, Poland
- 14.02.2019 “Generalised manifolds as basic objects of General Relativity”
Thursday Seminars, Oxford, United Kingdom
- 28.05.2018 “Quantum correlations in composite systems under global unitary operations”
Chaos and Quantum Information, Jagiellonian University, Kraków, Poland
- 27.11.2017 “Nonlocality, entanglement and negative conditional entropy of two-qubit systems”
Chaos and Quantum Information, Jagiellonian University, Kraków, Poland

EDUCATION – PARTICIPATION IN SUMMER SCHOOLS

- 26.06.-1.07.2022 *International Summer Institute in Philosophy of Physics on the Philosophy of Cosmology and the Philosophy of Quantum Gravity*, Morzine, France
- 23-27.09.2019 *Kavli RISE Summer School on Gravitational Waves*, Cambridge, UK

23-30.07.2018	<i>Particle Physics at the crossroads</i> , Wuppertal, Germany
16-21.07.2018	<i>Split Summer School in the Philosophy of Physics: The Chimera of Entropy</i> , Split, Croatia
25-28.06.2018	<i>XXI International Summer School in Philosophy of Physics: Philosophy of QFT</i> , Urbino, Italy
31.07.-4.08.2017	<i>6th Tuebingen Summer School in the History and Philosophy of Science, Understanding Relativity Theory: Special and General</i> , Tuebingen, Germany
11-18.03.2017	<i>Rethinking Foundations of Physics</i> , Dorfgastein, Austria
15-25.07.2015	<i>Cosmology School: Introduction to Cosmology</i> , Kielce, Poland

ACADEMIC SERVICE

2020	Co-editing <i>Modality in Physics</i> – Special Issue of <i>Foundations of Physics</i> (with Gábor Hofer-Szabó and Tomasz Placek).
11-13.06.2018	Co-organising a conference <i>Modality in Physics</i> , Kraków, Poland.
9-10.06.2017	Co-organising a conference <i>Modally Rich Metaphysical Landscapes</i> , Kraków, Poland.
27-28.10.2016	Co-organising a conference <i>Konferencja Filozofii Nauki i Metod Formalnych w Filozofii</i> [Conference on Philosophy of Science and Formal Methods of Philosophy], Kraków, Poland.

SERVICE AS A REFEREE

Referee for philosophical and physical journals: *Analiza i egzystencja* (1), *Axiomathes* (1), *The British Journal for the Philosophy of Science* (1), *Diametros* (1), *International Journal of Theoretical Physics* (1), *Philosophy of Science* (3), *Studies in History and Philosophy of Science* (1), *Zagadnienia filozoficzne w nauce* (2).

TEACHING EXPERIENCE

- Classes for undergraduate obligatory courses for Philosophy students: *Wprowadzenie do filozofii* [Introduction to Philosophy], *Epistemologia I* [Epistemology I];
- Classes for a graduate obligatory course for Philosophy students *Epistemologia II* [Epistemology II];
- Classes for obligatory course for Cognitive Science students *Epistemologia* [Epistemology];
- Lectures for undergraduate obligatory course for Psychology students *Wprowadzenie do filozofii* [Introduction to Philosophy];
- Seminars for philosophy students: *Filozoficzne implikacje mechaniki kwantowej* [Philosophical implications of quantum mechanics], *Wielkie metafizyczne pytania w fizyce* [Big metaphysical questions in physics].

POPULARIZATION OF PHILOSOPHY

2023	Popularizing paper about the possibility of relating the concepts of necessity and accident to physics: “Przypadek i konieczność w fizyce?”, <i>Filozofuj!</i> 53(5).
2022	Popularizing paper about the reality of the wave function: “Czy funkcja falowa reprezentuje coś w świecie?”, <i>Filozofuj!</i> 43(1), pp. 12-14.
2018-2019	Content editor for 3/2019 issue of polish journal popularizing philosophy <i>Filozofuj!</i> ,

entitled “Tajemnicza przyczynowość” [“The mysterious causality”].